causalgeo: concurrent geo holdouts for bundle lift and calibrated channel responses

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Abstract

Marketing teams often turn off different bundles of channels in different groups of markets at the same time. A geo holdout then gives the lift of each bundle, but it cannot say how much of that lift belongs to each channel. A marketing-mix model can make the split, though the split is unstable when channels move together. `causalgeo` is a Python package for this design. It assigns markets to cells within groups of similar markets, estimates one lift per bundle with uncertainty computed by market, and checks whether a channel meant to stay on delivered differently in the dark markets. It then fits a straight-line model of channel impressions that is pulled toward the measured bundle lifts, and it leaves a channel blank when the cells and impressions cannot separate it from the others. We test the package on Google's public Meridian impression paths with constructed sales, so the true responses are known. The recovered responses are within 0.21 percent of the planted values. On a separate 30-week series, the 95 percent cell intervals covered the true lift in 91.8 percent of 200 redrawn assignments. The checks also show where the split fails. A response that differs by market lowers coverage to 34.4 percent, and leaving out carryover or saturation gives large channel errors and a large gap between the model and the holdout.

Keywords: geo experiment, incrementality, holdout, marketing mix, Python

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Table 1: Code metadata.

Nr.	Code metadata description	Value
C1	Current code version	0.2.3
C2	Permanent link to code/repository used for this code version	`https://github.com/adityapt/causalgeo/releases/tag/v0.2.3`
C3	Permanent link to Reproducible Capsule	`https://doi.org/10.5281/zenodo.23787152`
C4	Legal Code License	MIT
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python
C7	Compilation requirements, operating environments and dependencies	Python $\geq$ 3.10. Runtime dependencies: pandas, numpy, scipy, statsmodels. The example also uses matplotlib.
C8	Link to developer documentation	`https://github.com/adityapt/causalgeo/blob/main/README.md`
C9	Support email for questions	puttaparthy.aditya@gmail.com

1. Motivation and significance

A geo test estimates incrementality by comparing markets that receive a media change with similar markets left on business as usual [1, 2]. Comparing markets avoids the bias that comes from lining up people who saw an ad with people who did not [3, 4], and it asks whether the channel created incremental sales [5].

In practice only some markets can go dark. Consider a United States campaign across 210 designated market areas (DMAs) where the business allows a pause in only 30. Large markets may be too risky to pause, some regions may have partner commitments, and others may already be in a flight. Advertising lift is small next to ordinary sales movement [6], so a well-powered test of a single channel would use the whole pool. Teams therefore test bundles of channels in several cells at once, and each cell gives the lift of its bundle and says nothing about the channels inside it.

GeoLift and Meridian GeoX can design and estimate more than one cell [7, 8]. Neither splits a cell into channels or ties a marketing-mix model to the measured cell lift. A mix model estimates channel response from impression history, and when channels move together that split is arbitrary [9, 10]. Robyn and PyMC-Marketing can pull a mix model toward a lift test, including a test of more than one channel [11, 12]. Pulling toward a test can steady a split [13], but it cannot add information that the test and the impression data do not contain. The split in this paper also assumes that the bundle lift is the sum of the channel lifts. If channels interact, these equations cannot identify the separate responses [14].

`causalgeo` estimates the bundle lift from the holdout and then uses that lift to guide a straight-line model of channel impressions. The channel responses therefore come from both the test and the model, and they are not channel-level holdout estimates. When the impressions cannot separate the channels, the package reports the bundle and stops. It does not supply a sales-per-impression weight of its own. It also does not control for in-flight spend, since going dark is what changes that spend [15].

2. Software description

2.1. Software architecture

The functions `design_concurrent_test` and `disentangle` are built from four modules.

1. `design.py` assigns markets within groups of similar markets, builds the cell-by-channel flight schedule, and records which channels can be split.
2. `identified.py` estimates one bundle lift per cell. It also estimates a channel lift when that channel was the only one turned off in a cell.
3. `exposure.py` checks whether a channel meant to stay on delivered differently in the dark markets than in the business-as-usual markets.
4. `calibrated.py` fits the impression response and uses each measured bundle lift to guide the channel split.

The bundle estimate removes each market's own baseline and each week's common movement, and it repeats that removal when a market is missing some weeks. Uncertainty is computed by market, so the weeks within one market are treated as one unit and not as separate observations [16, 17, 18].

The example has 40 markets with 8 dark markets in each cell. On a separate 30-week series of that size, we redrew the assignment 200 times, and the stated 95 percent intervals covered the true cell lift 91.8 percent of the time. For a single cell that is already assigned, `analyze` can also run time-based regression [19], synthetic control [20, 21], or synthetic difference-in-differences [22]. The synthetic-control interval comes from the pre-test residual spread alone and is too narrow for inference, so the package warns when it is used. Market selection and the power search are left to GeoLift and GeoX [7, 8].

2.2. Software functionalities

Design. The user supplies the markets, the groups of similar markets, the channel list, and the channels that go dark in each cell. Within each group, a fraction `control_share` of markets stays on business as usual and the rest are split across cells. If a cell map already exists, `design_from_assignment` accepts it with the same cell labels. The schedule sets $A_{kc} = 1$ when cell k turns channel c off. Each channel is then labeled not tested, tied, separate, or not separable. A channel that the flight never turns off is labeled not tested and gets no response.

Bundle lift. $D_{it}^{(k)} = 1$ when market i is in cell k and week t is in the test window. The cell estimate is

$$Y_{it} = \sum_{k} \tau_k \, D_{it}^{(k)} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

τ_k is the incremental sales of cell k after removing each market's baseline and each week's common movement. It is a channel estimate only when the cell turned off a single channel. Overlapping bundles can cover every channel, yet the lift still belongs to the whole bundle and cannot be assigned to any one channel in it.

Channels left on. A channel left on in cell k can still deliver differently in the dark markets than in the business-as-usual markets. The impression gap is

$$S_{kc} = \big(\bar{X}_{c,k,\text{post}} - \bar{X}_{c,k,\text{pre}}\big) - \big(\bar{X}_{c,\text{control},\text{post}} - \bar{X}_{c,\text{control},\text{pre}}\big). \tag{2}$$

The pre-test weight b_c is the joint response of the channels and is not an estimate of sales for one channel. The published cell lift remains τ_k. Suppose $|b_c S_{kc}|$ is larger than `material` times the larger of $|\tau_k|$ and the standard deviation of pre-test sales across all markets and weeks. The channel split

then uses a refit that accounts for those still-on impressions. The value of `material` is 0.10 unless the user changes it, and the user can choose to publish the refit as the cell lift. Channels that were turned off form the test itself and are never used as controls. The example table shows the requested refit.

Channel split. W_{kc} is the impressions of channel c held out of cell k,

$$W_{kc} = \max(-S_{kc},\, 0) \tag{3}$$

for the channels that the cell turned off. Every other entry in that row is zero. The mix model is asked to make $\sum_c \beta_c W_{kc}$ close to $-\tau_k$, where β_c is sales per impression. Each β_c is held at zero when the fit would otherwise slope down. The bundle lift τ_k is not held at zero. This row is a weighted pull toward the lift test and does not act as a hard constraint. The impression history uses every pre-period week, but during the flight it uses only the business-as-usual markets, so dark market-weeks are left out. A more precise bundle lift pulls harder. The bundle table shows whatever gap remains.

The reported uncertainty treats the cell lifts as separate. It does not fully account for shared control markets, or for weeks that enter more than one estimate. The simulations check this shortcut on the design used here and on no other flight.

Two impression paths whose absolute cosine similarity exceeds $1 - \epsilon$ are treated as copies, where ϵ is `copy_tolerance` and defaults to 10^{-6}. One copy stays in the fit so that their shared delivery is still accounted for, and neither copy gets its own response. By default a response is marked unstable if it moves when the calibration pull is weakened, or when the market with the largest in-flight sales is left out. The tables below switch this check off, so they show point estimates. A blank impression stays blank, the market-day is dropped, and the fit issues a warning. Daily impact is the delivered impressions times $\hat{\beta}_c$. The package can apply geometric adstock. It does not fit a saturation curve, because a single on/off holdout cannot trace a response curve [23, 10].

One bundle. One cell measures a single number, the bundle lift τ_k. The held-out impressions W_{kc} tell us how many impressions of each channel were removed, but not how many incremental sales each removed impression was worth. Call that unknown weight p_c, in sales per impression. The holdout gives one equation,

$$\sum_c p_c W_{kc} = -\tau_k. \tag{4}$$

With several channels in the bundle there are more unknowns than equations, and the package does not estimate p_c.

A user who already has p_c, from an older single-channel test or from a mix model they are willing to stand behind, can split the measured lift as

$$a_{kc} = \tau_k \frac{p_c W_{kc}}{\sum_j p_j W_{kj}}. \tag{5}$$

The channel pieces add up to the cell lift. Setting every p_c equal to 1 splits the bundle by held-out impressions, and any other weight is an assumption the user brings. We tested this on the public Meridian impression paths with all five channels dark together and sales built from known weights. A split by impressions gives Channel2 7.8 percent of the bundle against a true share of 2.4 percent, and Channel1 12.7 percent against 19.2 percent. Putting the known weights into (5) closes both gaps. This check uses the same weights that built the sales, so the holdout itself does not discover them. The script is `examples/public_allocation_benchmark.py`.

The multi-cell fit described earlier is a different calculation. Because different cells turn off different channels, the impression paths can separate the responses, and when they cannot the response stays blank. Equation (5) is a labeled split of one measured lift and is not a channel incrementality estimate.

The output has the channel responses, the measured lift next to the lift implied by those responses, the still-on impression check, and daily impact. A market bootstrap reruns the whole pipeline on markets redrawn within each cell.

3. Illustrative examples

No public geo holdout comes with a known channel-level incrementality. The example therefore uses Google's published Meridian geo file `geo_all_channels.csv`, which has 40 markets and 156 weeks from 25 January 2021 through 15 January 2024 [24]. The impressions come from that file and the sales are constructed so that the channel responses are known. The example checks the software and says nothing about whether a campaign worked. Meridian's conversions column is used only to group similar markets. It is not used as the outcome because that series already contains Meridian's own media response.

Table 2: Cell lift on the constructed series, with the still-on refit requested. By default the unadjusted holdout stays the published number and the impression shift is only recorded. The model column values the dark-channel impression change at the fitted responses.

Cell	Bundle	Unadjusted	Adjusted	SE	Model
1	Channel0+Channel1	−555.7	−555.7	62.9	−570.2
2	Channel0+Channel2	−318.0	−251.5	36.2	−253.6
3	Channel1+Channel2	−399.9	−399.9	54.4	−396.2

We use three of the five channels, with planted responses 2×10^{-4}, 5×10^{-4}, and 1×10^{-4} sales per impression. Markets are grouped into quartiles of average conversions. With `control_share = 0.4` and seed 1, 16 markets stay on business as usual and 8 go into each cell. The flight is the last 16 weeks, starting 2 October 2023. In a dark market-week the turned-off channel's impressions are set to zero, and sales are

$$Y_{it} = \alpha_i + \gamma_t + \sum_c \beta_c X_{itc} + \varepsilon_{it}, \tag{6}$$

with $\alpha_i \sim N(0, 50^2)$, $\gamma_t \sim N(0, 20^2)$, and $\varepsilon_{it} \sim N(0, 30^2)$, drawn with seed 7. The script is `paper/example.py`.

The three cells go dark on {Channel0, Channel1}, {Channel0, Channel2}, and {Channel1, Channel2}. No channel is ever dark alone, so a holdout-only table leaves all three channel estimates blank. Table 3 reports the responses from the impression model after it is pulled toward the three measured bundle lifts.

This checks only whether the package recovers the straight-line response it was built to fit. A real flight with a smaller lift will have a wider interval [6].

Cell 2 shows the still-on check. Channel1 was meant to stay on, but its impression gap, valued at the joint pre-period response, is worth −59 sales. That is 19 percent of the unadjusted lift and 14 percent of the pre-test sales standard deviation, which is the larger scale and above the 10 percent threshold. Refitting moves the estimate from −318.0 to −251.5. Cells 1 and 3 keep the unadjusted holdout.

Table 2 and Figure 1 put the holdout lift next to the lift implied by the channel responses. The gaps are 2.6, 0.8, and 0.9 percent of the holdout lifts, and every model total lies inside the holdout interval. The bundle number

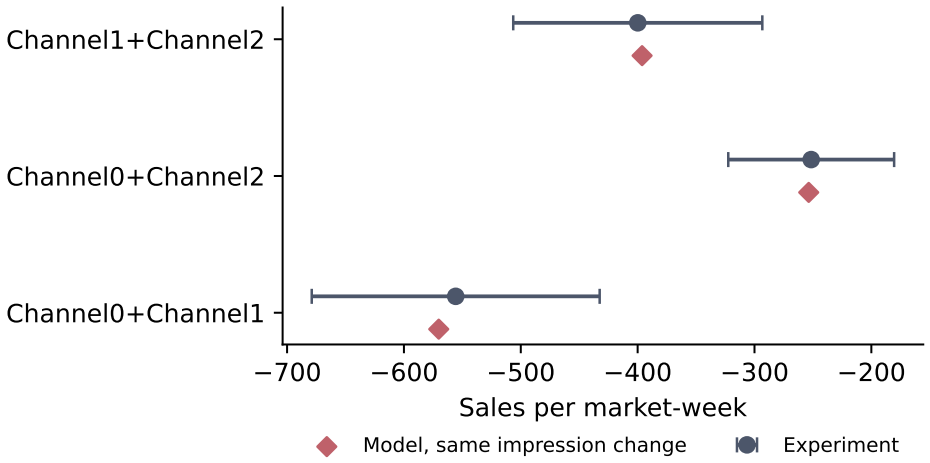


Figure 1: Measured bundle lift, with a 95 percent interval, and the same dark-channel impression change valued at the fitted responses. Sales per market-week.

Table 3: Planted and recovered response, in sales per 10,000 impressions. The largest absolute error is 0.21 percent.

Channel	Planted	Recovered	SE	Abs. error (%)
Channel0	2.000	1.997	0.006	0.17
Channel1	5.000	5.009	0.008	0.17
Channel2	1.000	1.002	0.008	0.21

comes from the holdout, and the channel numbers come from the model after it is pulled toward that holdout.

The same three cell lifts and the held-out impressions can also be solved for the three channel responses without any pre-period impression history. That holdout-only split gives 2.22, 4.48, and 2.01 sales per 10,000 impressions, with uncertainty 0.43, 0.64, and 1.14. The calibrated uncertainty on the same series is 0.006, 0.008, and 0.008. The tighter intervals therefore come from the impression history and not from the three cell lifts.

When Channel1 is replaced with a copy of Channel0, both channels stay blank. Their shared path remains in the fit as one control, and Channel2 comes back as 1.00 sales per 10,000 impressions, matching the planted 1.00. If the shared path is dropped instead, Channel2 rises to 3.03. Figure 2 shows both fits against the planted response.

Table 4: Two hundred refits per setting. Rho is how far the live Channel1 impression path is mixed toward Channel0, and the correlation is what that mix produces. Noise multiplies the example's leftover sales movement. Error is the median absolute percent error among reported responses. The brackets are the range of that coverage across the simulated tests. At rho 1, error and coverage refer to the channel that is still separate.

Noise	Rho	Corr.	Runs	Error (%)	Coverage [MC interval]	Withheld
1	0.0	0.34	200	0.19	0.958 [0.942, 0.974]	0.00
1	0.9	0.96	200	0.40	0.953 [0.934, 0.973]	0.00
1	1.0	1.00	200	0.58	0.940 [0.907, 0.973]	1.00
4	0.0	0.34	200	0.76	0.958 [0.942, 0.974]	0.00
4	0.9	0.96	200	1.62	0.957 [0.938, 0.975]	0.00
4	1.0	1.00	200	2.25	0.940 [0.907, 0.973]	1.00



Figure 2: Channel2 when Channel1 is an exact copy of Channel0. Keeping the shared delivery as one control recovers the planted response. Dropping that path pushes the remaining response from about 1 to about 3 sales per 10,000 impressions.

Table 4 and Figure 3 summarize 200 draws for each mix and noise level. At the example noise, the median absolute error is 0.19 percent when the impression paths are distinct (correlation 0.34) and 0.40 percent when the mix weight is 0.9 (correlation 0.96). The 95 percent intervals cover the truth 95.8 and 95.3 percent of the time. When one channel is an exact copy of another, the pair is left blank in every run and the remaining channel is covered in 94.0 percent of runs. Multiplying the leftover sales noise by four raises the error for distinct paths from 0.19 to 0.76 percent, while coverage stays at 95.8 percent. These results hold for this design and for uncertainty

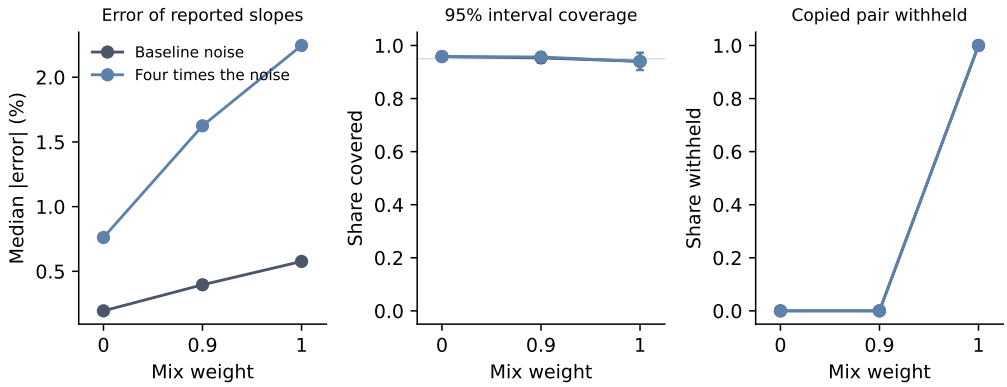


Figure 3: Recovery over 200 simulations as the live Channel1 impression path is mixed toward Channel0. The second series multiplies leftover sales noise by four. The bars are the range of coverage across those simulations. At mix weight 1 the pair is left blank, so error and coverage refer to the channel that is still separate.

that treats the cells as separate.

Table 5: Stress test over 200 draws when the sales response is not the straight line the package fits. Error is the median absolute percent error in each channel's contribution to the holdout contrast. Coverage is how often the stated 95 percent interval contains that contribution. The brackets are the range of that coverage across the draws. Gap is the median absolute difference between the model bundle and the holdout lift, as a percent of the holdout lift.

Fit	Runs	Error (%)	Coverage [MC interval]	Gap (%)
Carryover omitted	200	47.27	0.000 [0.000, 0.000]	48.09
Carryover modeled	200	0.19	0.942 [0.922, 0.961]	2.22
Saturation omitted	200	26.99	0.000 [0.000, 0.000]	26.97

The recovery check above uses the same straight-line response that built the sales. Table 5 instead turns on two responses that the fit can miss, over 200 draws. Carryover keeps half of last week's impressions in this week's effect. Saturation uses $\beta_c X/(1+X/h_c)$, where h_c is twice the median positive delivery and β_c is the response at the first impression. Neither response has a single true sales-per-impression number, so the error is judged against each channel's known contribution to the same dark-versus-BAU contrast as the holdout. The carryover case is also refit with the decay that built the sales.

With that decay included, the error returns to the straight-line result. With carryover or saturation left out, the channel errors are large and the 95 percent intervals miss the truth in every draw. The bundle gap is also large, so the package exposes the disagreement and does not let a bad channel split pass as a clean lift estimate.

Table 6: Further checks, 30 draws each, so the coverage ranges are wide. Error is the median absolute percent error among reported responses. The confounded row reports the Channel0 error. The brackets are the range of coverage across the draws. Serial correlation means this week's leftover sales still look like last week's, with correlation 0.8. Market-specific response lets each market sit between half and one and a half times the planted response. Redistribution moves half of a typical held-out week onto a channel the cell left on. Incomplete delivery leaves 30 percent of the business-as-usual level still running. The confounder follows Channel0's usual delivery and does not go dark. The collinear rows mix Channel1 toward Channel0 with weight 0.999, correlation 0.9995, and responses at 5 percent of the example.

Setting	Runs	Error (%)	Coverage [MC interval]	Gap (%)
Serial correlation	30	0.43	0.956 [0.914, 0.997]	3.56
Heterogeneous slopes	30	4.81	0.344 [0.248, 0.441]	4.01
Budget redistributed	30	0.22	0.967 [0.930, 1.000]	12.60
Redistribution, refit requested	30	0.22	0.967 [0.930, 1.000]	12.60
Incomplete delivery	30	0.22	0.967 [0.930, 1.000]	2.83
Observational fit only	30	0.22	0.967 [0.930, 1.000]	2.12
Calibrated fit	30	0.22	0.967 [0.930, 1.000]	2.13
Confounded Channel0	30	49.94	0.644 [0.614, 0.675]	21.12
Collinear, observational	30	30.94	0.833 [0.748, 0.919]	37.73
Collinear, calibrated	30	10.52	0.989 [0.967, 1.000]	8.79
Experiment only	30	11.16	1.000 [1.000, 1.000]	–

Table 6 asks what that recovery depends on. Sales that carry over from week to week, and a recorded shortfall in delivery, leave the channel error close to the straight-line result. A response that differs by market does change it, and coverage falls to 34.4 percent. When lost impressions are moved onto a channel that was left on, the channel error stays near 0.2 percent. The raw cell lift sits 52.9 percent away from the lift after those extra impressions are removed, because the holdout still contains the moved delivery. The channel split is calibrated to the cleaned lift, and the remaining bundle gap is 12.6 percent. Publishing that refit leaves the split unchanged, so the two budget rows agree. On the original series the impression-history fit and the calibrated fit both have 0.22 percent error.

A confounder that tracks Channel0 and stays on during the holdout leaves that response 49.9 percent high after calibration, against 50.0 percent from the impression history alone. The bundle gap opens to 21 percent while the response itself barely changes. Calibration does change the split when the channels are nearly copies and the planted responses are small. At a mix weight of 0.999 the impression-history error is 30.9 percent and the calibrated error is 10.5 percent, and the bundle gap falls from 37.7 percent to 8.8 percent. Both coverage ranges come from only 30 draws and are wide. The holdout-only split on the original series has a median error of 11.2 percent and coverage of 1, because its intervals are wide. Figure 4 shows coverage and the bundle gap alongside the carryover and saturation checks.

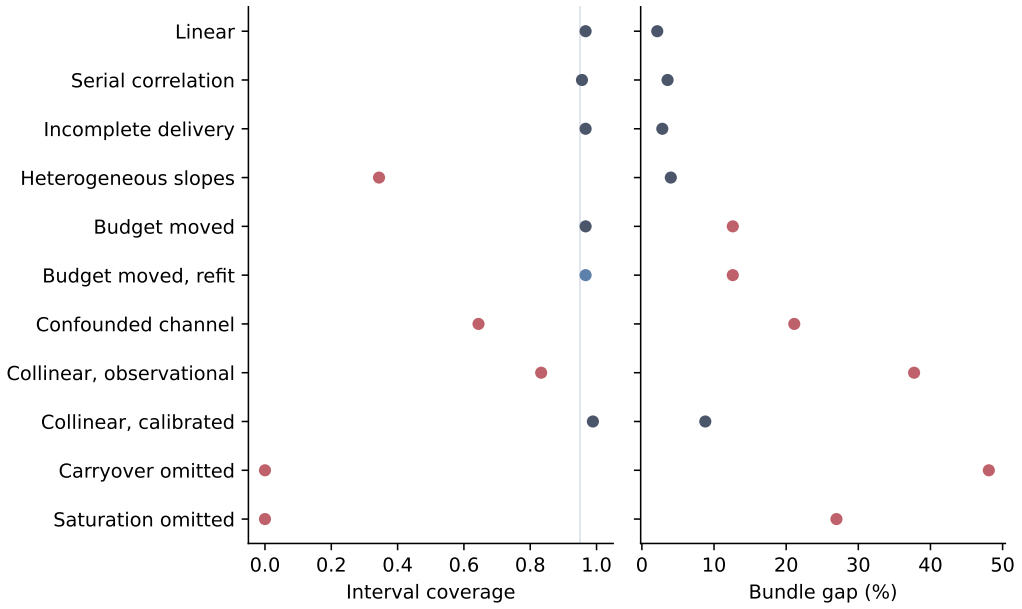


Figure 4: Coverage and the bundle gap. The straight-line, serial, delivery, market-response, budget, confounder, and collinear rows are 30 draws. Carryover and saturation are the 200-draw checks in Table 5. The pale line is 95 percent coverage.

4. Impact

As described in Section 1, GeoLift, GeoX, Robyn and PyMC-Marketing cover parts of this problem [7, 8, 11, 12]. `causalgeo` adds the concurrent holdout, a report of the gap between the holdout lift and the model, and a

blank result for any channel that the impressions cannot separate. Splitting a single bundle with a sales-per-impression weight rests on an assumption the user supplies, and the package does not estimate that weight.

The current release is 0.2.3. Adoption is early and this paper includes no user study. The script `paper/reproduce.py` rebuilds all five tables and four figures. The same functions apply to a real holdout once the user supplies market outcomes and channel impressions. Channels that the flight cannot separate, or whose impressions are copies, stay blank and receive no response.

5. Conclusions

`causalgeo` estimates concurrent channel-bundle holdouts and splits a measured bundle only when the impression paths can separate the channels. A single bundled test measures the lift of that bundle. The impressions removed from each channel are observed, but the sales per impression are not. Equation (4) is one equation for all the channels in the bundle, so the channel estimates stay blank unless other cells separate them. Equation (5) splits the measured lift once the user supplies the sales-per-impression weights, which are an input and cannot be estimated from the holdout. The calibration pulls the mix model toward the lift test and still shows the gap that remains.

On one constructed series the responses are within 0.21 percent of the planted values, and the calibrated split is far tighter than a holdout-only split. The checks also show where the split fails. A response that differs by market cuts coverage to 34.4 percent, a confounder that the holdout leaves on keeps the affected response about 50 percent high, and leaving out carryover or saturation produces large channel errors and large bundle gaps. All of these intervals depend on the response that was fitted and on treating the cells as separate.

The split is a straight line, with each slope held at zero when it would otherwise be negative, optional geometric adstock, and no saturation curve [10]. Turning a channel on or off does not trace the response curve, so the package does not fit one, and it does not estimate an interaction between channels [14]. Market selection and power are left to GeoLift and GeoX, and the package makes no claim that eight markets give a powered cell [7, 8, 6]. In-flight spend is left out of the estimate because the assignment is what changes it [15, 3].

CRediT authorship contribution statement

Aditya Puttaparthi Tirumala: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review and editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The impression paths are Google's public Meridian file `geo_all_channels.csv` [24]. `examples/meridian_conversions.py` runs the same split on Meridian's conversions column. That run has no known channel incrementality and only shows the software working on that file. The archived v0.2.2 release [25] has the DOI `https://doi.org/10.5281/zenodo.23787152`. In `paper/`, set `PYTHONPATH` to `../src` and run `python3.11 reproduce.py`. That rebuilds Tables 2–6 and the four figures, then checks them against the rounded values reported here. The `--fast` option only checks that the scripts finish. Table 1 gives the repository links.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Cursor, with the Grok language model, to draft and revise the manuscript and the example code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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